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Status of occupational health and safety of waste workers in Chattogram city and the associated challenges

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Abstract

This study examines the occupational health and safety (OHS) status and challenges faced by waste management and sanitation workers in Chattogram City, Bangladesh. Despite the critical role these workers play in urban cleanliness, there is a lack of empirical research focusing on their specific OHS challenges in this context, particularly regarding gender-specific issues and the effectiveness of current safety measures. This study aims to address this gap by evaluating workers' OHS knowledge, attitudes, and practices, including the use of personal protective equipment (PPE), hygiene habits, and willingness to seek medical treatment and OHS training, and by providing recommendations to improve occupational health and safety practices among waste workers. A social survey was conducted using semi-structured interviews with 200 randomly selected workers. These workers were between 18 and 52 years old. Participants came from a variety of occupations, including sanitary pit collectors and road sweepers. The findings revealed poor hygiene practices among the workers and highlighted specific challenges faced by women waste workers, including discrimination and a lack of childcare services. The study found a high prevalence of chronic diseases among waste workers, with 64.0% reporting muscle pain, 46.3% experiencing back pain, and 22.2% suffering from asthma. Furthermore, access to hygiene facilities and financial assistance was limited; only 8% of workers had access to government-provided healthcare. Despite the significant health risks, only 18% of workers received any OHS training. A notable association between occupation and asthma, aligning with previous research on factors affecting cardiorespiratory health, was also observed. These findings underscore the urgent need for targeted policies and interventions in waste management. The study provides recommendations to improve occupational health and safety practices among waste workers, policymakers, employers, and civil society to help protect the health and rights of waste and sanitation workers in Bangladesh.

Keywords Occupational health and safety, Chronic diseases, Hygiene practices, Waste workers



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1 Introduction

Workers in waste management play a crucial role in keeping any community clean and safe. These employees oversee the collection, transport, and disposal of waste and garbage from homes, businesses, and public locations, including hazardous materials [7]. Their profession, however, comes with several occupational health and safety dangers, along with other possibilities of physical injuries and exposure to toxic fumes that can lead to death [19]. Workers in waste sectors face various physical, chemical, biological, and ecological dangers that can have serious short- and long-term health consequences [14]. Typical risks include toxic chemicals, biological contamination, noise, heavy lifting, and working in confined places [4]. Despite the critical nature of their work, they often receive minimal recognition and inadequate protection, leading to severe health consequences.

In many developing countries, waste workers operate under informal conditions with limited protective equipment and access to health services [22]. These workers often belong to marginalized communities and have low socioeconomic status, which intensifies their vulnerability. Studies have shown that waste workers are frequently exposed to hazardous materials, including sharp objects, toxic substances, and biological waste, which can lead to injuries and infections [6, 8, 15, 18]. The lack of proper training exacerbates these risks, as many workers are unaware of the best practices for handling waste safely. In addition to physical hazards, waste workers face significant psychological stress. The stigmatization and social exclusion associated with their work can lead to mental health issues, including anxiety and depression [12]. Furthermore, waste workers often have limited access to healthcare services, which delays the treatment of occupational injuries and illnesses, leading to chronic health problems and diminished quality of life.

The occupational health and safety (OHS) of waste workers is an area that requires urgent attention. According to the International Labour Organization (ILO), waste collection is one of the most dangerous jobs globally, with workers facing risks from chemical exposure, ergonomic stress, and mechanical hazards. Despite these significant risks, OHS regulations and policies often overlook waste workers. Many countries lack comprehensive frameworks that address the specific needs of waste workers, leaving them unprotected against occupational hazards [2, 11]. Moreover, waste management companies rarely provide adequate safety training or protective equipment to their employees, particularly in the informal sector. To mitigate these risks, effective occupational health and safety policies that target the specific hazards faced by waste and sanitation workers are required [10]. The situation is significantly worse for female workers. They face challenges of unhygienic working conditions, discrimination, lack of maternity leave, and weekly holidays [1]. Waste workers often lack awareness of personal hygiene and fail to practice it at work [21]. Female waste workers experience additional vulnerabilities, including workplace discrimination, sexual harassment, lack of adequate sanitation facilities designed for women's needs, and difficulties balancing demanding work schedules with family responsibilities and childcare obligations.

Bangladeshi sanitation workers face several health risks from various hazardous elements and a lack of PPE, with challenges including limited training and inadequate hygiene facilities [17]. Workers in waste and sanitation are exposed to infections and have significant potential to transmit diseases to others since they operate in residential

areas, on public roads, in hospitals, and in other establishments [20]. The primary reasons for the high rate of infections among workers have long been identified as inadequate PPEs, lack of standard operating procedures, lack of hygiene knowledge and occupational health safety training, and inadequate hygiene facilities at the workplace ([3, 16]). Currently, the 41 wards of Chattogram City Corporation have a total of 3,072 employees working as waste and sanitation workers. The city has a growth rate of 2.49%, and it is expected that by 2023, there will be approximately 5,380,000 inhabitants in the greater Chattogram Area. On average, each ward has a population of 131,220 people, but there is only one waste and sanitation worker for every 1,752 individuals. This staffing ratio indicates a substantial workload burden and suggests potential understaffing, which may contribute to occupational health and safety risks. While the general occupational hazards faced by waste workers have been documented in various international and national studies, there is a notable lack of empirical research focusing specifically on the unique challenges and context of waste management and sanitation workers in Chattogram City, Bangladesh. In particular, little is known about the gender-specific challenges, the effectiveness of current OHS practices, and the lived experiences of these workers within this urban setting. This research gap is especially pronounced in Bangladesh, where rapid urbanization and population growth have significantly increased the demand for waste management services while institutional capacity and regulatory frameworks have struggled to keep pace with these demands.

This study addresses this critical knowledge gap by providing context-specific evidence on OHS knowledge, practices, and health outcomes among waste workers in Chattogram City and by offering targeted policy recommendations to improve their working conditions. The study aims to investigate the occupational health and safety challenges faced by waste management and sanitation workers in Chattogram City by analyzing survey data and conducting qualitative interviews and individual discussions. It focuses on demographic characteristics, prevalent health issues, occupational health and safety training attendance, and financial safety incentives. Specifically, this study seeks to address the lack of empirical research on OHS challenges in this context, with particular attention to gender-specific issues and the effectiveness of current safety measures. The objectives are to evaluate workers OHS knowledge, attitudes, and practices, including the use of PPE, hygiene habits, and willingness to seek medical treatment and OHS training; assess the prevalence of chronic diseases and health conditions among different occupational categories, examine gender-specific challenges and barriers faced by female waste workers, and analyze the relationship between occupational factors and health outcomes. By integrating both quantitative and qualitative insights, this study aims to generate evidence-based recommendations for policymakers, employers, and civil society to protect the health and rights of waste and sanitation workers in Bangladesh and similar developing country contexts.

2 Methodology

2.1 Study location

Chattogram is the second-largest metropolitan city and also the commercial capital of Bangladesh, mainly located in the southeastern part of the country. The study area is situated between longitudes 91°40' and 91°53' east and latitudes 22°13' and 22°27' north (Fig. 1). The 170 square kilometer Chattogram City Corporation is divided into

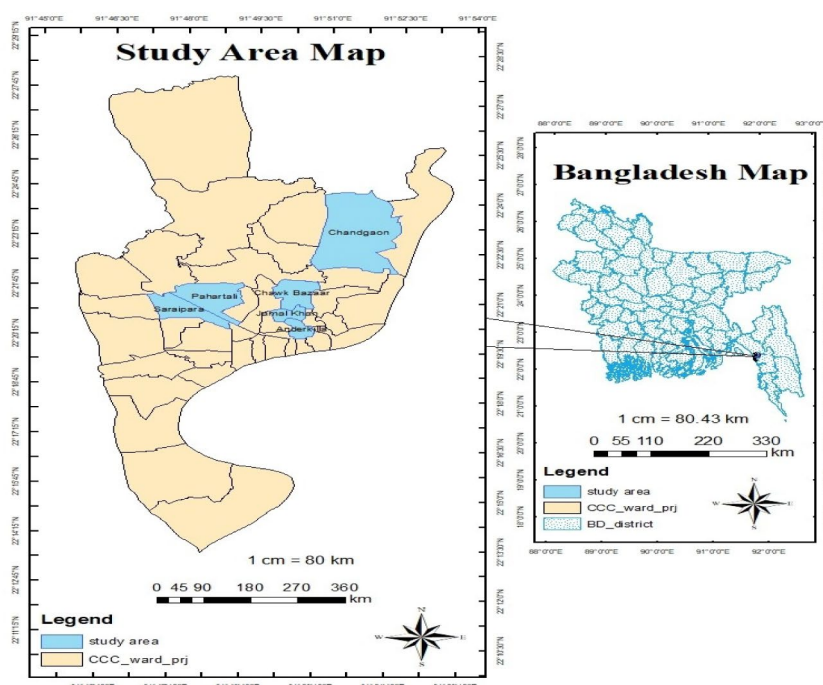


Fig. 1 Map of the study area

41 wards, with the cleanliness of each ward being primarily maintained by waste and sanitation workers. Its population density is 32,008 people per sq km. The 41 wards of Chattogram City Corporation currently employ 3072 waste and sanitation workers. The Chattogram metro area is expected to be home to 5,380,000 inhabitants in 2023, with the city's growth rate currently at 2.49. On average, each ward contains 131,220 residents, and there is just one waste and sanitation worker for every 1752 people. Waste and sanitation workers range from 18 to 52 years old.

2.2 Study design and sampling methods

2.2.1 Sampling and participants

The target population for the quantitative survey was all waste management and sanitation workers across the 41 wards, totaling 3,072 individuals. Using the Yamane formula with a 95% confidence level and a 5% margin of error, the calculated sample size was 384. However, due to operational constraints such as worker shift patterns and accessibility, 260 workers were approached, with 200 consenting and completing the survey, yielding a response rate of 76.9%. Simple random sampling was performed using official worker lists provided by Chattogram City Corporation. Eligible workers were assigned unique numbers within each ward and selected via a random number generator, ensuring representation across occupational categories (pit collectors, road sweepers, drain cleaners, door-to-door service workers). This study employed a cross-sectional survey design for the quantitative component, with data collected from consenting participants at a single time point between June and August 2023.

2.2.2 Data collection procedures for quantitative survey

Quantitative data were collected via interviewer-administered structured questionnaires at workers' respective worksites during breaks or rest periods to minimize disruption. The questionnaire comprised three main sections: socio-demographic and economic information, occupational health and safety status, and hygiene practices (Supplementary file). This was developed based on a thorough literature review and expert consultation, and underwent a reconnaissance survey to assess content validity and test-retest reliability (Cronbach's $\alpha=0.87$). The purpose of the reconnaissance survey was to gain a better understanding of the research areas and identify those that would be employed as study areas with high potential for producing results on this subject. The community survey aimed to understand the status of waste and sanitation workers in the study region. Accurate maps of the study area were crucial for the study. The map was produced using information from "Bangladesh Jorip Odhidoptor (Survey of Bangladesh)," along with additional sources such as articles, related works, journals, newspapers, magazines, and other publications. This study determined the frequency (%) of variables and their associations. Univariate analysis was employed to determine the frequency of socio-demographic features and other discrete variables within the research population. The importance of observed disparities in categorical variables was evaluated using the chi-square test, with a p-value of 0.05 or lower being significant. Interviews typically lasted 10–15 min per participant.

The questionnaire collects data on socio-demographic characteristics, including age, gender, employment status, and educational attainment (Table 1). The researchers designed the questionnaire following a comprehensive literature study and an examination of occupational health, safety, and hygiene practices among trash workers. The mean overall face validity was 95.

The survey for this study took place from June 2023 to August 2023, with only those present at the time of the survey responding.

The study protocol was approved by the Institutional Review Board (approved examination board) of the Department of Civil Engineering, Chittagong University of Engineering and Technology (CUET). The board reviewed and approved the study protocol, including the research design, data collection procedures, questionnaire instruments, and informed consent procedures, confirming that the study adequately protects participant rights, maintains confidentiality, and involves minimal risk to participants.

Table 1 Socio-demographic information for the survey

Section	Name of the section	Questions/variable
1	Socio-demographic and economic information	• Age group • Gender • Occupation • Education level • Accommodation system • Financial support
2	Occupational health and safety	• Common Chronic Diseases Among the Responders • Problems that female Workers Face in the Workplace • Participation in Any Occupational Training • Demand for Occupational Training
3	Hygiene Practices	• Hygiene Practice Status Among the Workers During Work • Hygiene Practice of the Waste Workers After Returning Home • Worker's Practice Reported to Prevent Infectious Diseases • Hand Washing Practice Among the Workers During Work

2.2.3 Data management and statistical analysis

Data were double-entered into IBM SPSS (version 28.0) by independent data clerks to ensure accuracy. Data cleaning involved range and consistency checks as well as exploration of missing data patterns. Descriptive statistics (frequencies, percentages, means, and standard deviations) summarized all variables.

Inferential analyses used chi-square tests of independence to evaluate associations between categorical variables such as occupation type and chronic disease outcomes, as well as demographic factors and hygiene practices. The choice of chi-square tests was made because both occupation and health outcomes were categorical, and thus more appropriate than association coefficients designed for continuous data. Effect sizes were interpreted using Cramer's V statistic. Statistical significance was set at $p \leq 0.05$.

Fifteen workers were purposively sampled from the quantitative survey participants to capture in-depth perspectives. Selection criteria included workers who reported chronic health issues, expressed concerns related to women's safety, or represented varied occupational roles. This purposive strategy enriched the data on nuanced, gender-specific, and occupational health challenges.

3 Results

3.1 Participant characteristics and employment conditions

A total of 200 waste workers participated in the quantitative survey, distributed across four occupational categories: door-to-door service workers (37%, $n = 74$), road sweepers (33%, $n = 66$), solid waste/pit collectors (17%, $n = 34$), and drain cleaners (13%, $n = 26$). Additionally, 15 workers participated in qualitative interviews to provide in-depth insights into their occupational experiences. The workforce was predominantly male (67%, $n = 134$), with female representation at 33% ($n = 66$). Most participants were middle-aged, with 41% aged 20–29 years, 34% aged 30–39 years, and 23% over 40 years. The religious composition of the workers showed 69% Hindu and 31% Muslim (Table 2).

Educational attainment was limited, with 38% of workers being illiterate, 47% completing primary education, and only 15% finishing secondary education. Employment conditions reflected significant economic vulnerability, with 93% of workers arranging their own accommodation and only 7% having access to government-provided housing. Financial support from authorities was minimal, with only 8% receiving any form of governmental assistance.

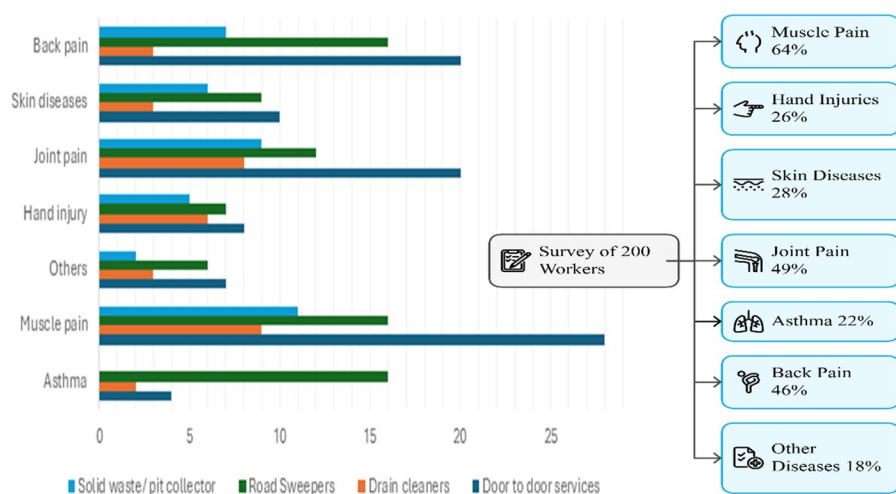
Workers typically worked 4–6 h daily despite being expected to complete 8-hour workdays. Monthly wages averaged 14,000 Taka (approximately \$120 USD), with only 8% receiving festival bonuses, indicating precarious employment conditions and limited social protection.

3.2 Health problem outcomes and healthcare access

The study revealed a high burden of occupational health problems among waste workers. Musculoskeletal disorders were most prevalent, with 64% reporting muscle pain, 49% experiencing joint pain, and 46% suffering from back pain. Respiratory conditions affected 22% of workers who reported asthma symptoms. Additionally, 28% experienced skin diseases, 26% suffered hand injuries, and 18% reported other health conditions. Figure 2 illustrates the distribution of common chronic diseases among different occupational categories with their respective percentages.

Table 2 Socio-demographic characteristics of study participants (N = 200)

Characteristic	Category	N	(%)
Gender	Male	134	67
	Female	66	33
Age	< 20	4	2
	20–29	82	41
	30–39	68	34
	> 40	46	23
Religion	Islam	62	31
	Hinduism	138	69
	Others	0	0
Education Level	Primary	94	47
	Secondary	30	15
	Illiterate	76	38
Occupation	Door-to-Door Service	74	37
	Solid Waste/Pit Collector	34	17
	Drain Cleaner	26	13
	Road Sweeper	66	33
	Others	0	0
Getting Financial Support from the Authority	Yes	16	8
	No	184	92
Accommodation Facilities	Government facility	14	7
	Own facility	186	93

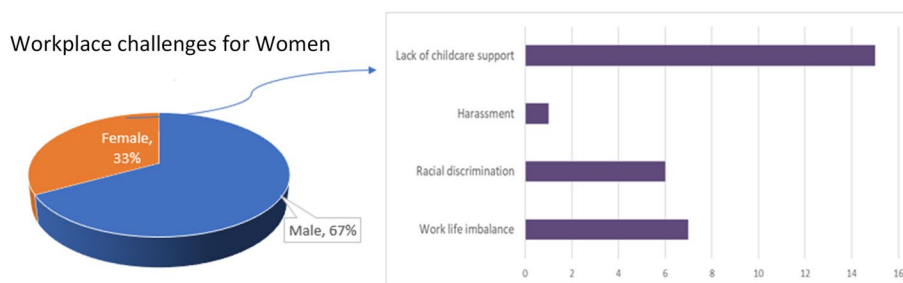
**Fig. 2** Common chronic diseases among different occupation with percentage

Healthcare access patterns revealed significant barriers to formal medical care. The overwhelming majority of workers (92%) relied on self-medication through local pharmacy shops, while only 6% accessed public hospitals during emergencies (Table 3). No workers reported using private healthcare services, and merely 2% had access to government-provided healthcare services.

Infectious disease prevalence was substantial, with 44.5% reporting gastrointestinal issues (diarrhea, dysentery), 33.5% experiencing skin infections (rashes, fungal infections), 28% suffering respiratory infections, including chronic cough, 17% reporting fever episodes potentially linked to vector-borne diseases, and 11.5% experiencing eye infections.

Table 3 Healthcare seeking behavior ($N = 200$)

Healthcare source	<i>n</i>	%
Local pharmacy (self-medication)	184	92.0
Public hospital (emergencies only)	12	6.0
Government-provided healthcare	4	2.0
Private healthcare	0	0.0

**Fig. 3** Problems which female workers face at the workplace ($N = 66$)

3.3 Occupational safety training, gender-specific challenges, and thematic findings

Occupational health and safety training coverage was critically inadequate, with only 18% of workers having received any form of OHS training. When surveyed about training needs, 43% believed no training was required, while 44% expressed demand for operation and maintenance training, 41% requested health and safety measures training, and 37% sought equipment handling training. Regarding training needs, 44% of workers expressed demand for operation and maintenance training, 41% requested health and safety measures training, and 37% sought equipment handling training.

Female waste workers faced distinct occupational challenges that compounded their vulnerability. Among female participants ($n = 66$), 81.8% reported work-life balance difficulties, 72.7% lacked adequate childcare facilities, 63.6% experienced workplace discrimination, 59.1% had inadequate sanitation facilities, and 50% expressed safety concerns. Figure 3 illustrates the comprehensive range of problems faced by female workers at their workplace.

The qualitative analysis of 15 interviews revealed six major themes: physical health burdens characterized by chronic pain and musculoskeletal disorders; inadequate safety measures including lack of personal protective equipment and insufficient training; gender-specific challenges encompassing facility inadequacies and safety concerns; social stigma and discrimination reflecting community attitudes and workplace exclusion; economic insecurity manifested through low wages and lack of benefits; and psychological stress involving anxiety and lack of recognition.

The following table summarizes the major themes and sub-themes identified from the qualitative data, accompanied by representative quotations from respondents that illustrate the subtle realities behind the statistical results (Table 4).

The thematic table presented above offers a concise synthesis of the core patterns and issues that emerged from the qualitative interviews with waste management and sanitation workers. By organizing the data into major themes and sub-themes, the table highlights not only the diversity of challenges faced by these workers but also the depth of their lived experiences. The inclusion of direct respondent quotations further illustrates the significance and authenticity of each theme, providing valuable context to the

Table 4 Thematic table: occupational health and safety challenges

Major theme	Sub-theme	Representative quotation
Physical Health Burdens	Chronic pain, musculoskeletal disorders	"We have to work long hours lifting heavy bins, and most of us have constant back pain." (R-1)
Inadequate Safety Measures	Lack of PPE, insufficient training	"They gave us gloves once last year, but they tore within a week. After that, nothing." (R-2)
Gender-Specific Challenges	Lack of facilities, safety concerns	"There are no separate toilets for women at our collection points." (R-5)
Social Stigma and Discrimination	Community attitudes, workplace exclusion	"People avoid us in public transport because of our work." (R-7)
Economic Insecurity	Low wages, lack of benefits	"If I take leave for sickness, my salary is cut." (R-3)
Psychological Stress	Anxiety, lack of recognition	"Sometimes I feel ashamed to tell others about my job." (R-8)

Table 5 Summary of hygiene practices among waste workers (N = 200)

Category of practice	Specific practice	Percentage (%)
Practices to Prevent Infectious Diseases	Hand washing	69.0
	Face mask use	53.0
	Washing shoes	17.0
	Washing work clothes	13.0
	No preventive measures	11.0
Hygiene Practices During Work	Use proper cough/sneeze etiquette	60.0
	Wear face masks	59.0
	Share personal items (e.g., cigarettes, cups)	36.0
	Maintain social distancing	12.0
Hand Washing Practices During Work	After using toilet	79.0
	Before meals	64.0
	Before and after work	44.0
	During work intervals	11.0
Hygiene Practices After Returning Home	Wash hands with soap	53.0
	Bathe with soap	52.0
	Wash work clothes	46.0
	Restrict family contact	15.0

statistical findings and reinforcing the study's mixed-methods approach. These qualitative insights are essential for understanding the nuanced realities behind the quantitative data and for informing more targeted and effective policy interventions.

3.4 Hygiene practices and infectious disease prevention

Personal hygiene and infection prevention practices varied considerably among workers. Hand washing was the most common preventive practice, reported by 69% of workers, followed by face mask use at 53%. However, only 17% washed their shoes regularly, 13% washed work clothes, and approximately 11% took no preventive measures. During work hours, 60% of workers used proper cough and sneeze etiquette, 59% wore face masks, though 36% shared personal items such as cigarettes, drinking water, or teacups, and only 12% maintained social distancing. Hand washing frequency analysis revealed that 79% of individuals washed their hands after using the toilet, 64% before meals, 44% before and after work, but only 11% during work intervals (Table 5).

Post-work hygiene practices showed that 53% washed hands with soap upon returning home, 52% bathed with soap, 46% washed work clothes, and 15% restricted family contact to prevent disease transmission.

Table 6 Association between occupation and chronic diseases

Health condition	χ^2	df	p-value	Cramer's V
Asthma	12.45	3	0.006*	0.249
Muscle pain	8.76	3	0.033*	0.209
Back pain	5.42	3	0.144	0.165
Joint pain	4.89	3	0.180	0.156
Skin diseases	2.34	3	0.505	0.108
Hand injuries	1.98	3	0.576	0.099

* $p < 0.05$ **Table 7** Statistical results of occupation and hygiene practice

Hygiene practice		All Workers (N = 200)	Door-to-door service Workers (N = 74)	Road sweepers (N = 66)	Solid waste/ pit collec- tors (N = 34)	Drain clean- ers (N = 26)
Practice to prevent infectious diseases	Face mask, Hand washing	33%	33%	22%	10%	9%
	Hand washing, alone	22%	-	22%	-	-
	Face mask, alone	10%	-	-	10%	-
	Washing shoes, Face mask	4%	4%	-	-	-
	Washing shoes, Hand washing	4%	4%	-	-	-
	Washing shoes, Face mask, Hand washing	1%	1%	-	-	-
	None	11%	-	-	-	9%

3.5 Statistical associations and hygiene practice patterns

Chi-square tests revealed significant associations between occupation type and specific health conditions. A statistically significant association was found between occupation and asthma ($\chi^2 = 12.45$, $df = 3$, $p = 0.006$, Cramer's $V = 0.249$), indicating a moderate effect size (Table 6). Similarly, occupation was significantly associated with muscle pain ($\chi^2 = 8.76$, $df = 3$, $p = 0.033$, Cramer's $V = 0.209$). However, associations between occupation and other chronic diseases, including back pain, joint pain, skin diseases, and hand injuries, were not statistically significant.

Analysis of training participation revealed no significant association between gender and receipt of OHS training ($\chi^2 = 2.15$, $df = 1$, $p = 0.143$), indicating that training deficits affected both male and female workers equally.

This study conducted a physical survey of waste and sanitation workers in four professions. It found that a maximum of 33% of workers wear face masks during work to prevent infectious diseases and practice hand washing regularly (Table 7).

4 Discussion

4.1 Occupational health burden and disease patterns

This cross-sectional study provides comprehensive empirical evidence on the occupational health and safety challenges faced by waste management and sanitation workers in Chattogram City, Bangladesh. The findings reveal a substantial health burden, with 64% of workers reporting muscle pain, 49% experiencing joint pain, and 46% suffering from back pain. These high prevalence rates of musculoskeletal disorders reflect the physically demanding nature of waste work, involving heavy lifting, repetitive motions, and prolonged standing, which aligns with previous research by [18] and [6] who documented similar patterns among municipal solid waste workers in developing countries.

The 22% prevalence of asthma among study participants is particularly concerning and significantly higher than the general population prevalence in Bangladesh. The statistically significant association between occupation type and asthma... indicates a relationship that may be explained by the varying exposure to dust and bioaerosols across different tasks, with roles such as sweeping and landfill operations potentially posing elevated respiratory risks. This finding corroborates the work of [8] and [15], who emphasized the respiratory hazards associated with waste exposure.

The substantial prevalence of infectious diseases, with 44.5% reporting gastrointestinal issues and 33.5% experiencing skin infections, reflects direct exposure to pathogens in waste materials. These findings are consistent with the WHO (2020) recommendations that identified elevated risks of infectious diseases among waste workers due to exposure to pathogens present in waste materials.

4.2 Healthcare access barriers, treatment patterns and gender-specific vulnerabilities

A critical finding is the overwhelming reliance on self-medication, with 92% of workers seeking care from local pharmacy shops rather than formal healthcare facilities. This pattern reflects significant barriers to healthcare access, including financial constraints, lack of knowledge about occupational health rights, and limited availability of services, as previously documented by WaterAid [20] and [17]. Only 2% of workers had access to government-provided healthcare, highlighting a substantial gap in occupational health services for this essential workforce.

The qualitative findings further illuminate these challenges, with workers reporting that salary deductions for sick leave discourage formal medical care seeking. This economic vulnerability perpetuates a cycle where occupational injuries and illnesses go untreated, potentially leading to chronic conditions and reduced work capacity over time, consistent with observations by [12].

Female waste workers face compounded challenges, with 81.8% reporting work-life balance difficulties, 72.7% lacking adequate childcare facilities, and 63.6% experiencing workplace discrimination. The absence of gender-appropriate sanitation facilities emerged as a particularly acute issue in qualitative interviews, reflecting broader systemic inequalities in waste management employment. These findings align with previous research by [1] and extend the understanding of gendered occupational health challenges in Bangladesh's informal waste sector. The intersection of gender, economic vulnerability, and occupational hazards creates multiple layers of risk for female workers. The lack of maternity leave and childcare support forces women to work in potentially hazardous conditions even during pregnancy and while managing family responsibilities, as documented in similar contexts by [12].

4.3 Training deficits and knowledge gaps, socioeconomic context, and systemic issues

The finding that only 18% of workers received any occupational health and safety training represents a critical gap in worker protection. This low training prevalence is consistent with research by [2] and [16], who documented similar patterns across Bangladesh. The fact that 43% of workers believed no training was necessary suggests limited awareness of occupational risks and the potential benefits of safety education.

Without proper knowledge of safe handling procedures, appropriate use of personal protective equipment, and emergency response protocols, workers remain vulnerable to

preventable injuries and illnesses. This aligns with findings from [11], who emphasized the role of training in reducing occupational hazards among informal waste workers.

The study's findings must be understood within Bangladesh's broader socioeconomic context. The monthly wage of 14,000 Taka (\$120 USD) represents significant income disparity, reflecting the marginalized status of waste workers. This economic vulnerability compounds health risks, as workers cannot afford to take sick leave or access private healthcare. The predominance of Hindu workers (69%) in waste management reflects historical occupational patterns, adding social stigma to economic marginalization. This intersection of religious identity, occupational status, and health risks creates unique vulnerabilities that require culturally sensitive interventions, as noted in previous research on waste worker discrimination [9].

4.4 Hygiene practices and infection control

While 69% of workers practiced regular hand washing, the limited availability of soap and sanitizers at worksites undermines infection prevention efforts. The finding that workers often purchase hygiene supplies at their own expense highlights the economic burden of maintaining occupational health and reflects inadequate employer support, as documented in similar studies by WaterAid [21].

The mixed hygiene practices observed, including 36% of workers sharing personal items and only 12% maintaining social distancing, indicate the need for comprehensive hygiene education and workplace policy implementation. These findings are consistent with previous literature highlighting inadequate hygiene facilities and limited OHS training as major contributing factors to high prevalence of infectious diseases among waste workers (Bank Group et al., n.d.; [16]).

The findings of this study underscore the significant occupational health and safety (OHS) challenges faced by waste management and sanitation workers in Chattogram City. The high prevalence of musculoskeletal disorders, respiratory illnesses, and poor hygiene practices observed among the respondents is consistent with previous research in both Bangladesh and other developing countries, which highlights the vulnerability of waste workers due to inadequate access to personal protective equipment, insufficient training, and limited healthcare services [17, 20]. The statistically significant associations identified between occupation and chronic diseases such as muscle pain and asthma further reflect the hazardous working conditions and continuous exposure to physical and environmental risks, corroborating the findings of [6]. Gender-specific challenges, including the lack of adequate sanitation facilities and the experience of workplace discrimination, were also prominent in this study and align with the observations of [1] and [12], who have emphasized the compounded vulnerabilities of female waste workers. The limited provision of OHS training and restricted access to government-supported healthcare, as reported by the participants, reinforce the urgent need for targeted interventions and policy reforms, echoing recommendations from both national and international studies. Collectively, these results highlight the global and local dimensions of OHS challenges in the waste management sector, emphasizing the need for context-specific, evidence-based strategies to safeguard the health and well-being of this essential workforce.

4.5 Policy implications and intervention strategies

The hierarchy of hazard controls suggests that effective interventions should prioritize systemic changes over individual protective measures [5]. While personal protective equipment provision is important, engineering controls (such as improved waste handling equipment), administrative controls (including rotation schedules and regular health monitoring), and hazard elimination strategies provide more comprehensive protection.

Priority interventions should include mandatory occupational health and safety training programs, establishment of workplace hygiene facilities, implementation of regular health screening programs, provision of adequate personal protective equipment, and development of gender-sensitive workplace policies. These measures are essential to safeguarding the health, safety, and dignity of waste management and sanitation workers, as emphasized by recent studies [13].

4.6 Study limitations and future research directions

Several limitations should be acknowledged. The cross-sectional design limits causal inference about occupational exposures and health outcomes. The focus on six wards within Chattogram City may limit generalizability to other urban areas or rural waste management contexts. Self-reported health data may be subject to recall bias, and the relatively small sample size for some subgroup analyses limits statistical power for detecting associations. Longitudinal studies are needed to establish causal relationships between occupational exposures and health outcomes among waste workers. Future research should expand to include rural waste management contexts and evaluate the effectiveness of specific interventions on worker health and safety outcomes. Additionally, research examining the economic impact of occupational health interventions could provide valuable evidence for policy advocacy. The findings underscore the urgent need for comprehensive occupational health and safety reforms in Bangladesh's waste management sector, requiring coordinated action from government agencies, employers, civil society organizations, and international development partners.

5 Conclusion

This study provides empirical evidence on the occupational health and safety (OHS) challenges faced by waste management and sanitation workers in Chattogram City, Bangladesh. The findings reveal a high prevalence of chronic diseases, such as muscle pain, back pain, and asthma, as well as inadequate access to hygiene facilities and government-provided healthcare among the workforce. The study also highlights poor hygiene practices and significant gender-specific challenges, including discrimination and insufficient childcare support for female workers. Furthermore, the results indicate that only a minority of workers receive OHS training, underscoring the need for more comprehensive and accessible educational programs. A key contribution of this study is the context-specific analysis of OHS challenges in Chattogram City, with a focus on both quantitative and qualitative dimensions, including gendered experiences and the effectiveness of current safety measures. However, the study is subject to certain limitations, including its cross-sectional design, reliance on self-reported data, and the focus on selected wards within Chattogram City, which may affect the generalizability of the findings. Future research should consider longitudinal studies to assess the long-term health

outcomes of waste workers and evaluate the effectiveness of specific interventions, as well as expand research to other urban and rural areas in Bangladesh. Based on the findings, it is recommended that policymakers and relevant stakeholders prioritize the provision of adequate personal protective equipment, regular OHS training, improved access to healthcare services, and the development of gender-sensitive workplace policies. These measures are essential to safeguarding the health, safety, and dignity of waste management and sanitation workers in Bangladesh, and the insights provided by this study offer a valuable foundation for evidence-based policy and further research in this critical area.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12982-026-01356-9>.

Supplementary material 1.

Author contributions

Tanveer Bin Fakhrul conducted a physical survey, writing a draft, Md. Jakaria Alam Reza: Conducted physical survey, writing draft, Md. Ajijur Rahman: writing draft, editing, revising. Mst. Dr. Farzana Rahman Zuthi: Supervise, conceptualize, and draft editing.

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Data availability

The datasets generated and analysed during the current study are not publicly available due to the sensitive nature of the information they contain, which could compromise the privacy of research participants, but are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with relevant ethical guidelines and regulations for research involving human participants. All methods were carried out in accordance with institutional and national guidelines for ethical research conduct. Ethical approval for this research was obtained from the Institutional Review Board (IRB), which is an approved examination board of the Department of Civil Engineering at Chittagong University of Engineering and Technology (CUET), prior to the commencement of data collection in June 2023. The review board of the institution reviewed social and professional ethics. They approved the study protocol, including research design, data collection procedures, questionnaire instruments, and informed consent procedures. The board confirmed that the study adequately protects participant rights, maintains confidentiality, and involves minimal risk to participants.

Informed consent

This study obtained informed consent prior to collecting data from participants. The surveyors communicated to the participants that the study involved minimal risk. They were provided information about the study's purpose and assured that participation was voluntary and that declining participation would not impact their employment status. Additionally, personal information was removed from the data to ensure anonymity, and confidentiality was guaranteed. Furthermore, it was emphasized that the study data would be used solely for research purposes.

Consent for publication

Not applicable. No individual or identifiable personal data is included in this study.

Competing interests

The authors declare no competing interests.

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